On Perfect Additive Randomized Encodings



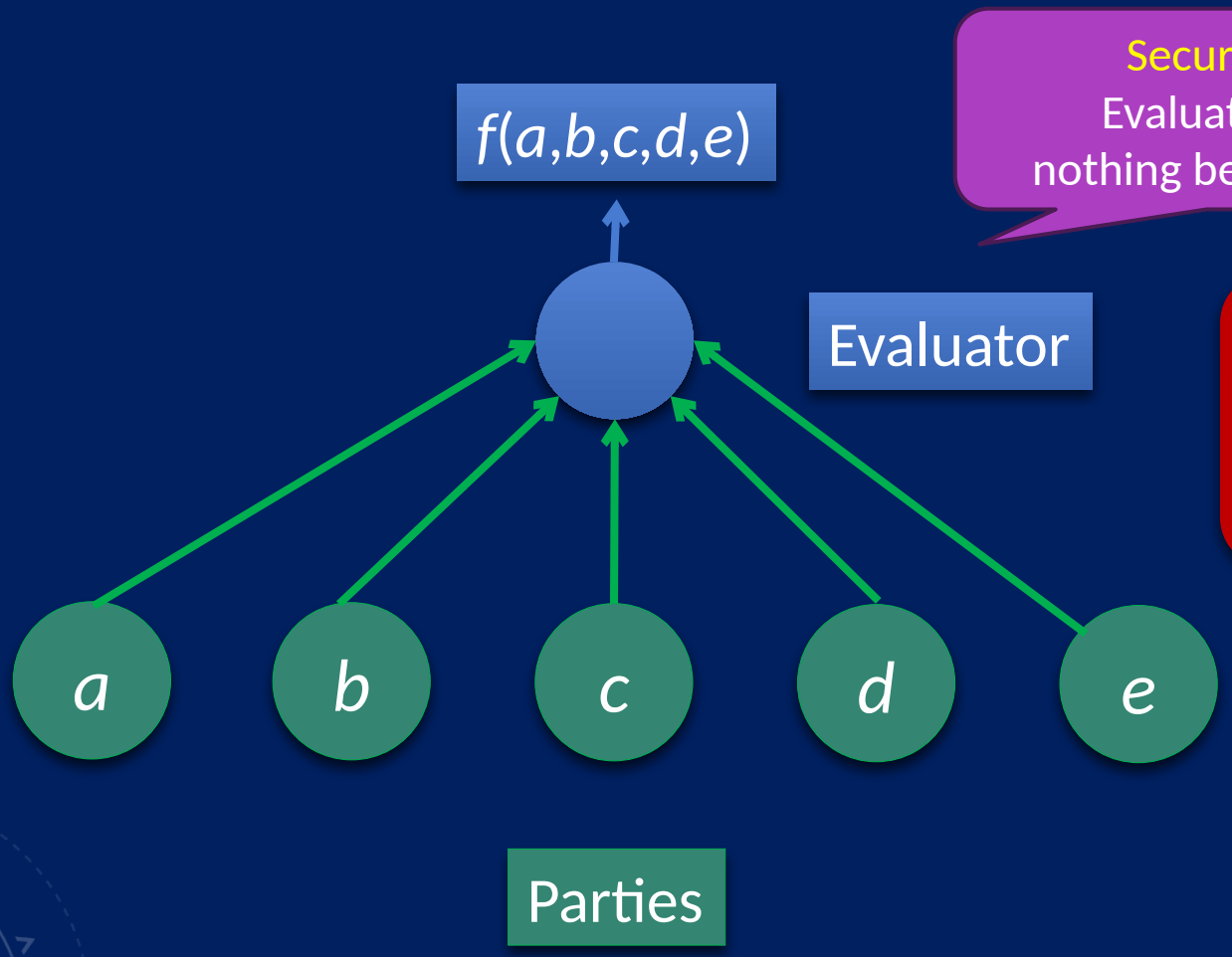
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November 10, 2026, TCC 2026

One-Round Secure Computation?

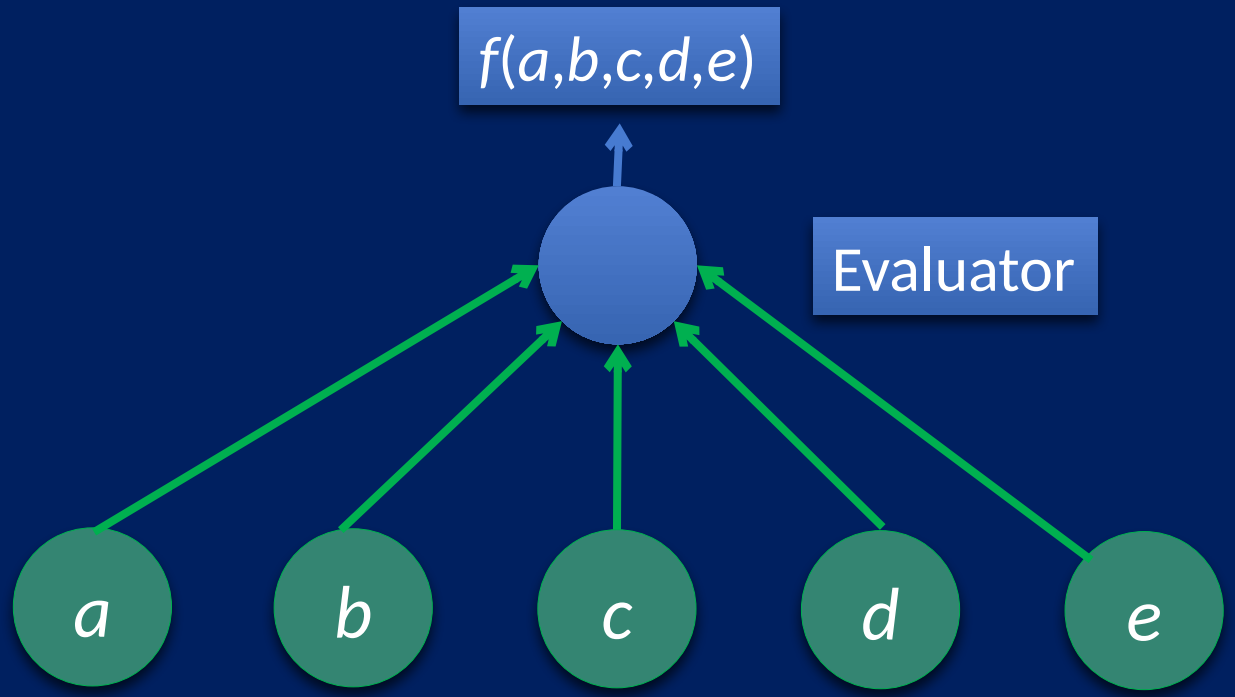


Security goal:
Evaluator learns
nothing beyond output

Impossible!

E.g., evaluator can compute
$f(0,0,1,1,e)$

One-Round Secure Computation with Anonymity?



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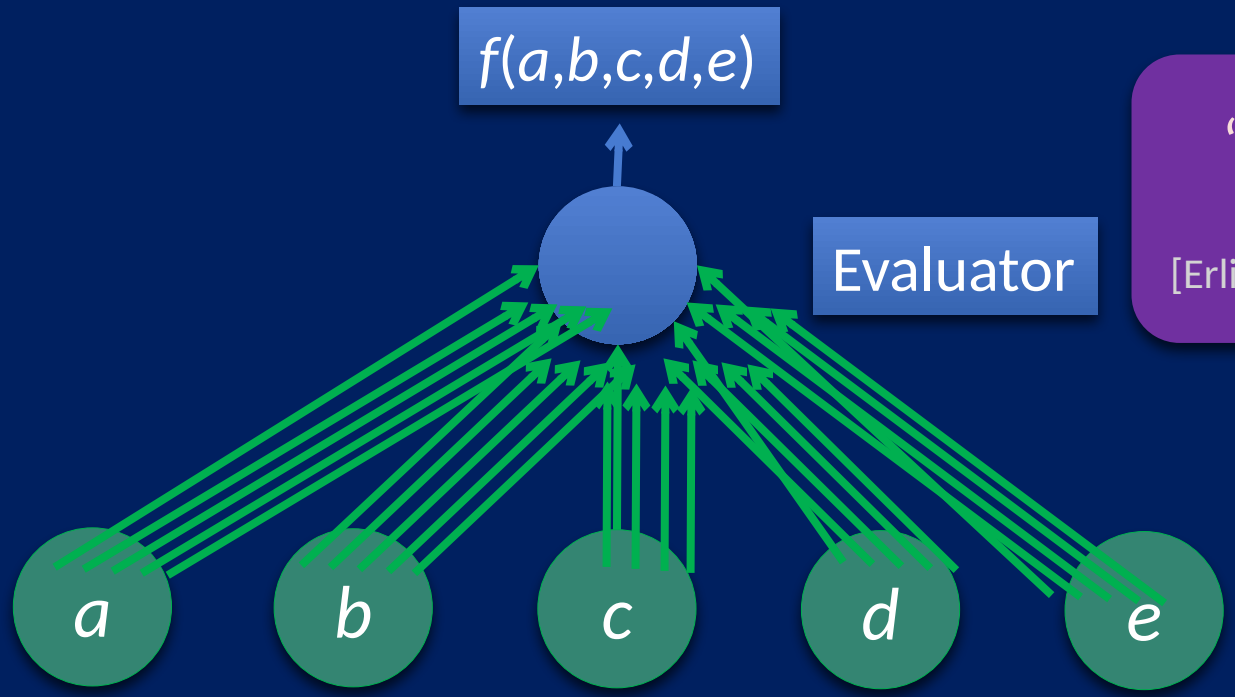
Still impossible…

E.g., for f = Majority
can learn Hamming weight

One-Round Secure Computation with Anonymity?

Cryptography from Anonymity
[I-Kushilevitz-Ostrovsky-Sahai 06]

"Shuffle Model" for Differential Privacy
[Cheu-Smith-Ullman-Zeber-Zhilyaev 19]
[Erlingsson-Feldman-Mironov-Raghunathan-Talwar-Thakurta 19]

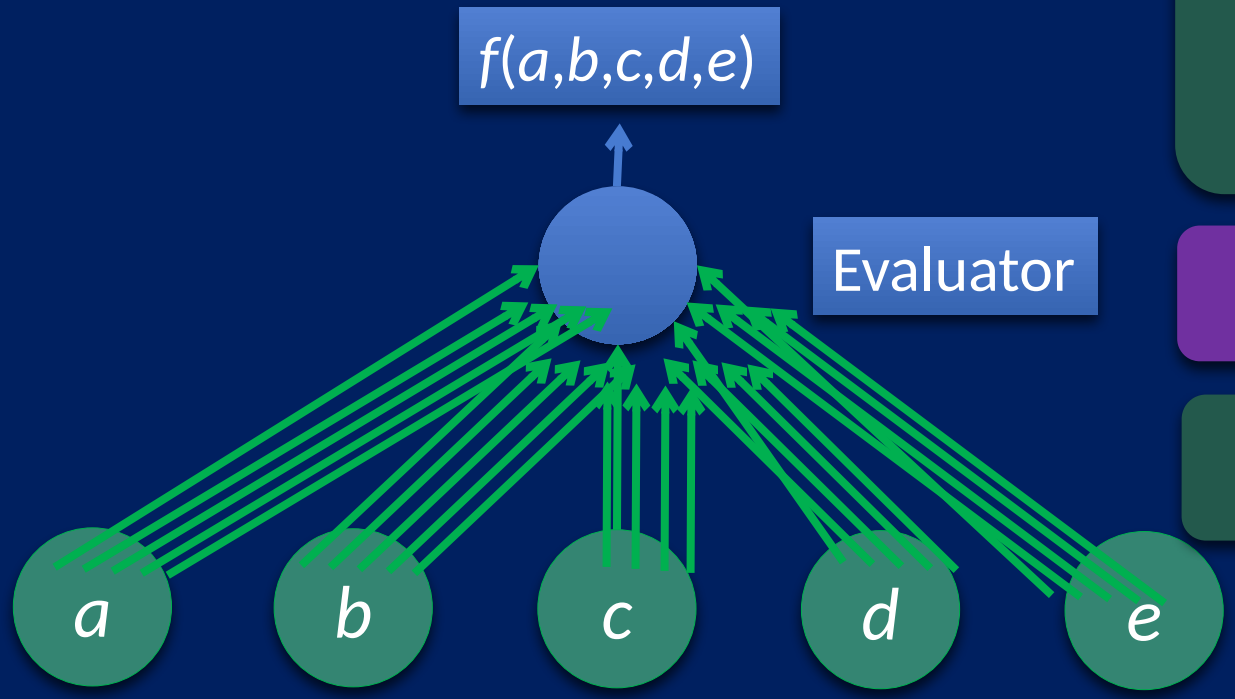


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One-Round Secure Computation with Anonymity?



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lkm3fs@gmail.com

ss35hd@gmail.com
qlkd9a@gmail.com
qpfll3@gmail.com
felt2d@gmail.com

ru3mdr
kex0e
fdsw
Fle

Attacks no longer apply!

Evaluator cannot guess the full set of messages of any party

But is it possible?

Yes for modular addition [IKOS06]

Additive Randomized Encoding

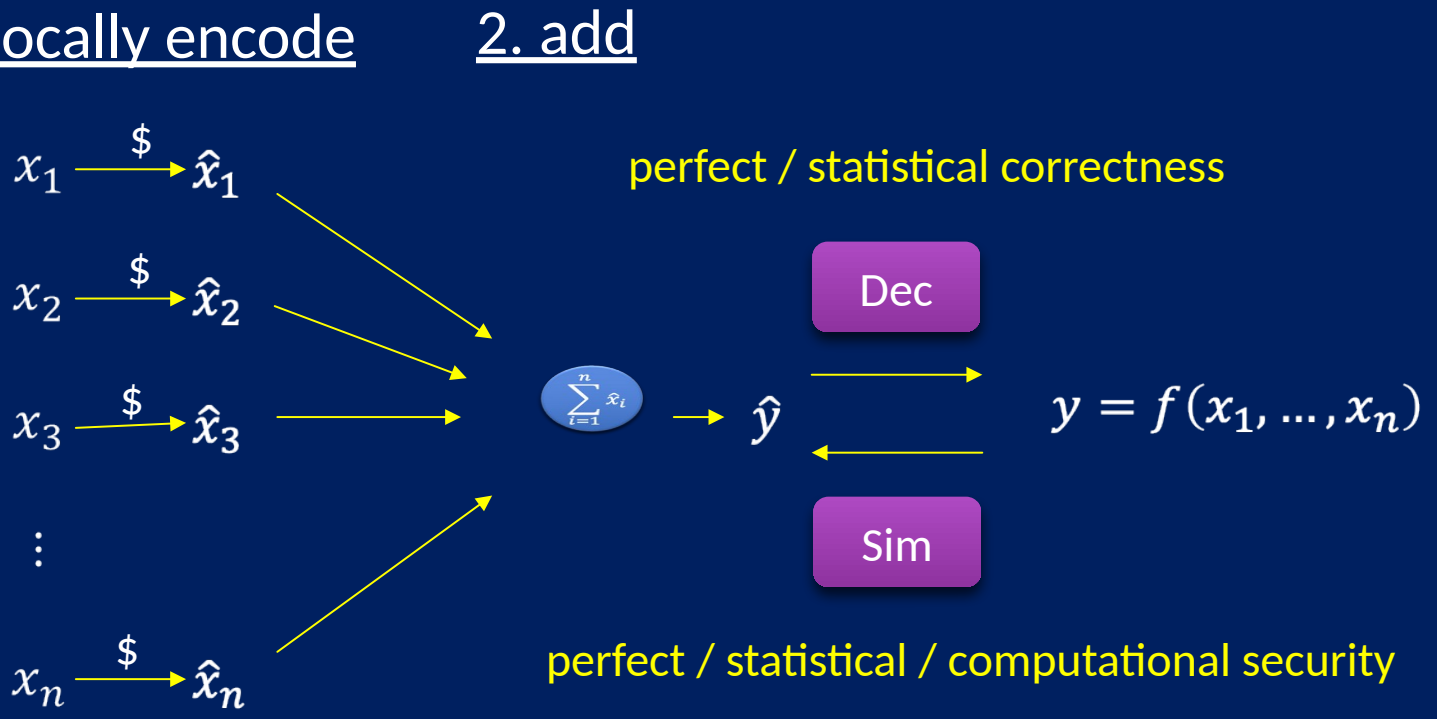
Yes in general! [Bitansky-Erabelli-Garg-I 26]
- Information-theoretic security
- Efficient in branching program size

Additive Randomized Encoding (ARE)

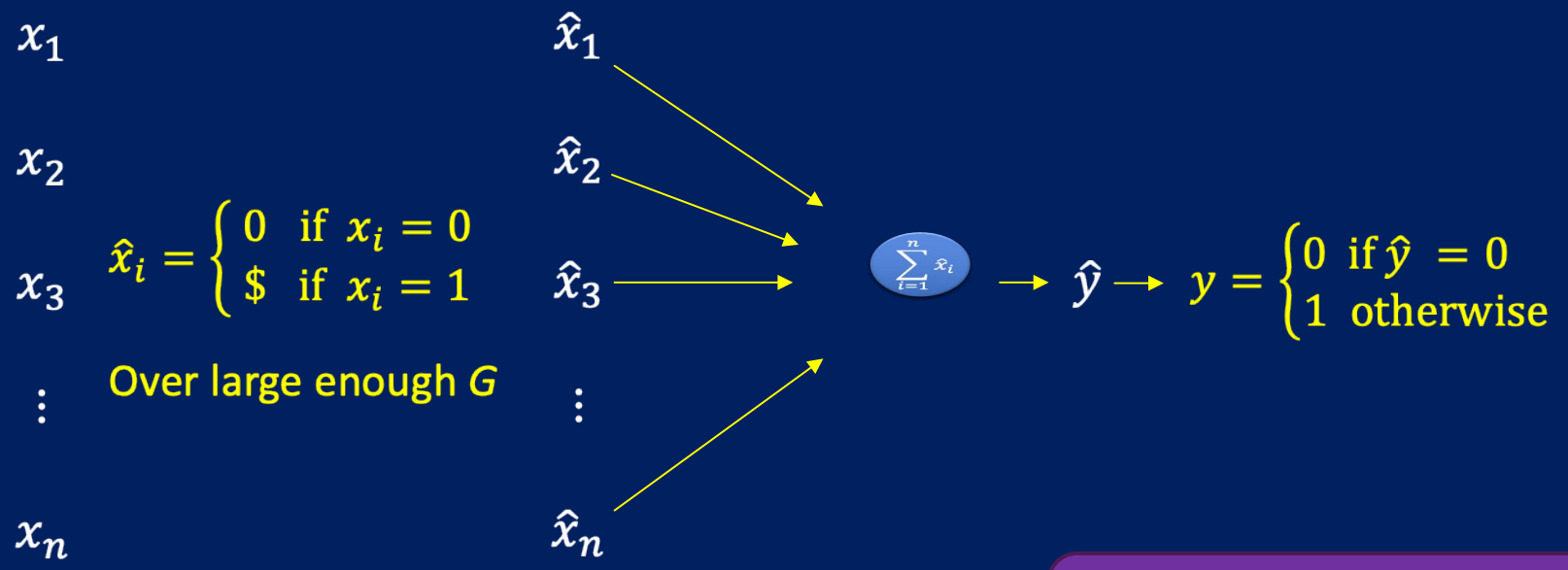
[Halevi-I-Kushilevitz-Rabin 23]

"Degree-1 RE with local preprocessing"

Fix an Abelian group $(G, +)$ (e.g., $G = \mathbb{Z}_p, G = \mathbb{Z}_2^k$)



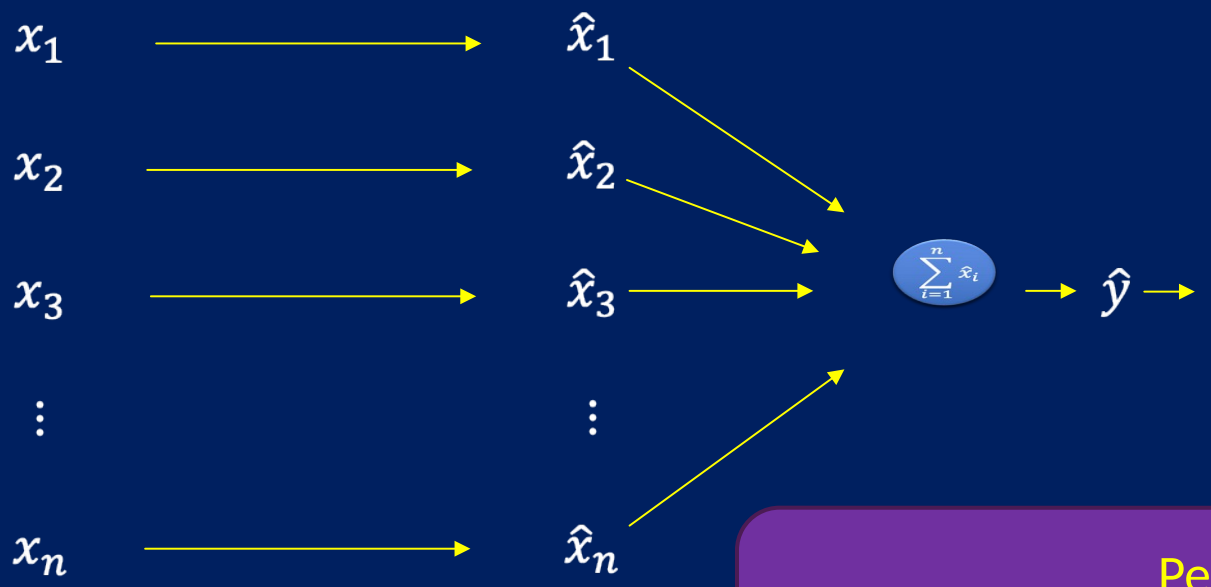
Perfectly Secure ARE for n-party OR



Can we eliminate it?

- Imperfect correctness: $1/|G|$ error
- Perfect security: $Sim(y) \rightarrow \{0 \text{ if } y = 0, \$ \text{ if } y = 1\}$

Statistical ARE for general functions [BEGI26]

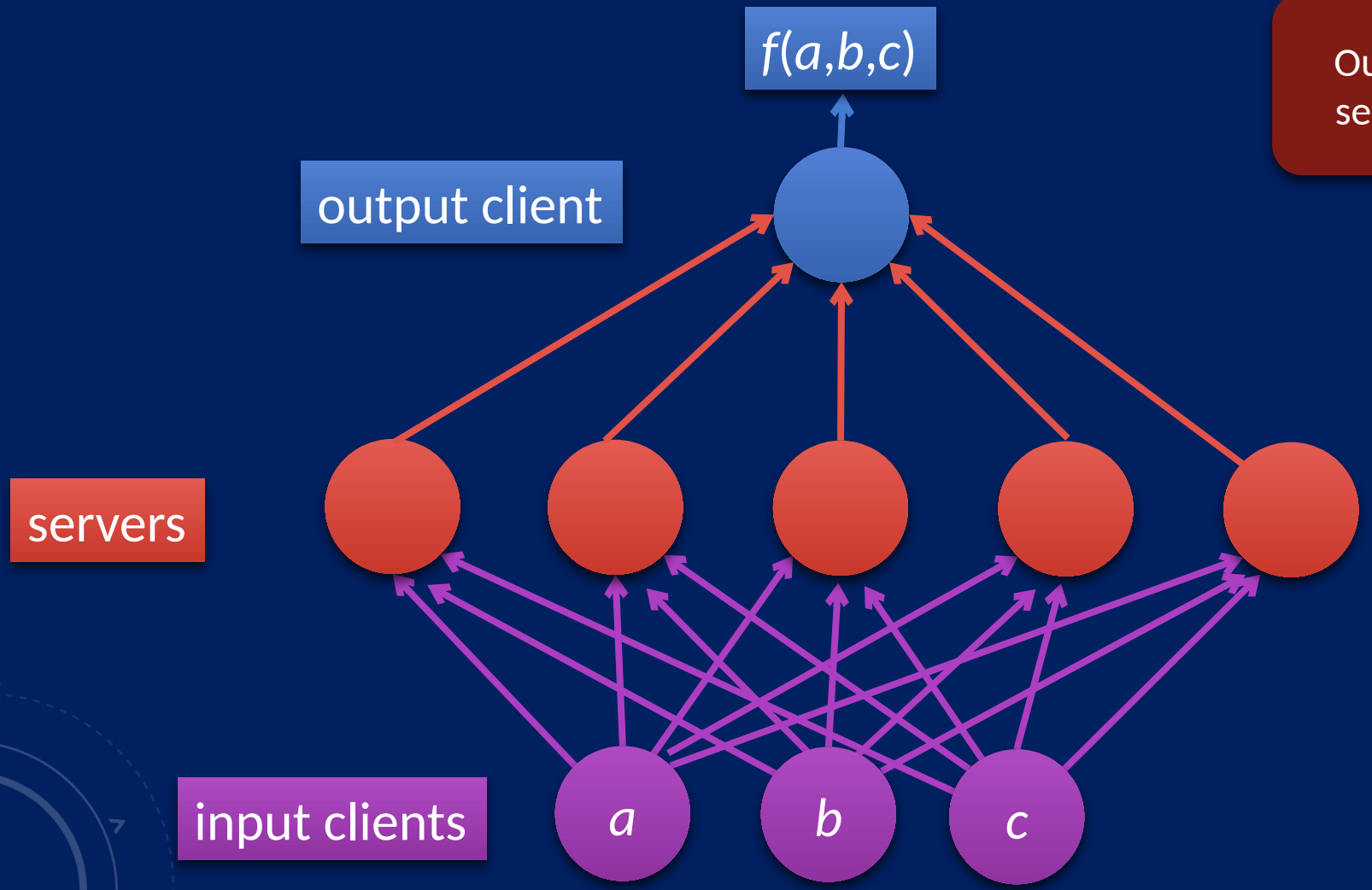


- Statistical correctness
- Statistical security

Perfect correctness?
Possible with computational security from PKE [BEG25]
or with “Las Vegas” relaxation [BEGI26]

Inherent for typical f [HIKR23]

Application: Client-Server MPC with Uncorrupted Inputs



Output client + any subset of servers (passively) corrupted

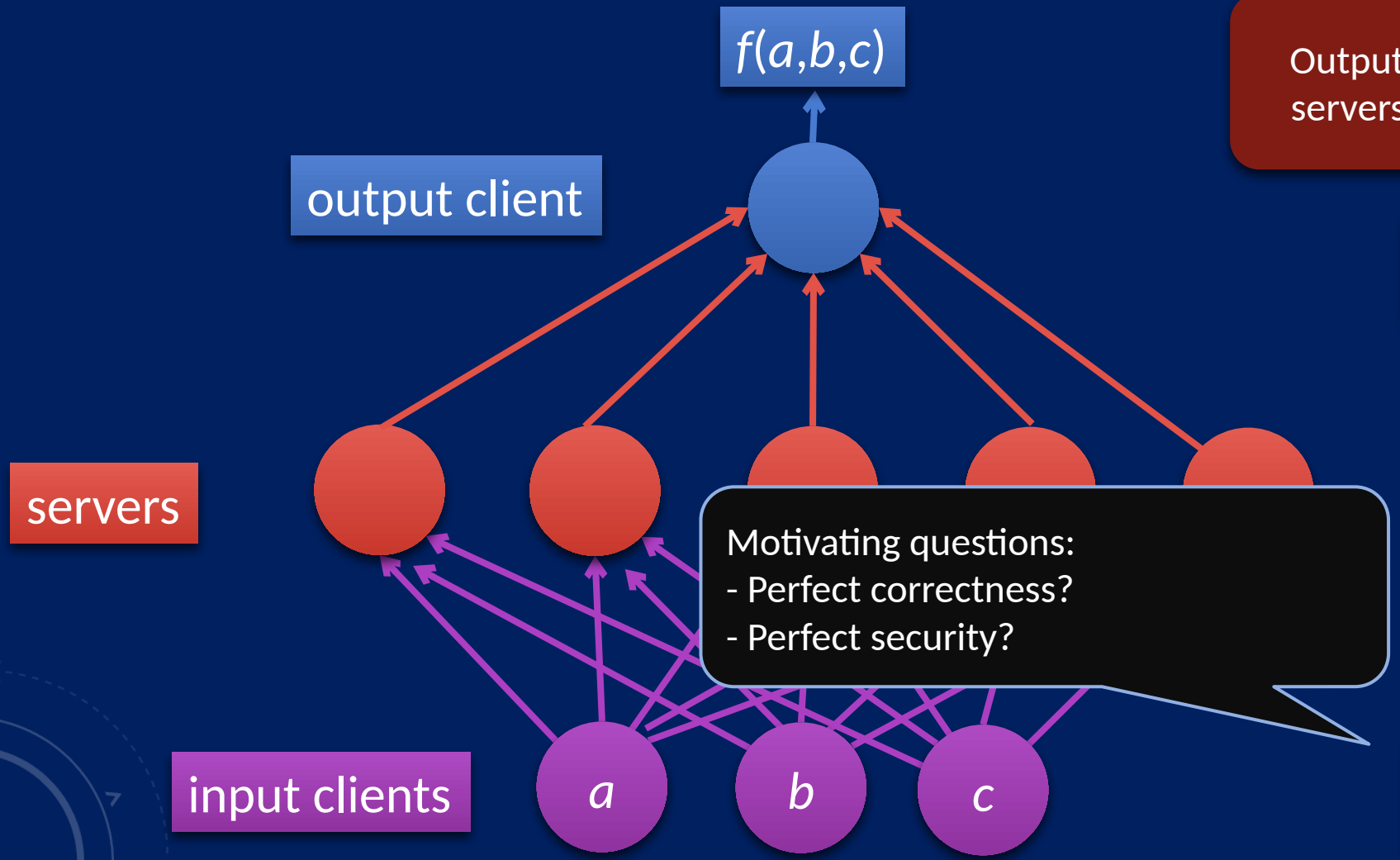
Perfect protocol for addition

For general f, computational security via (non-black-box use of) oblivious transfer

[Garg-Srinivasan 18, Benhamouda-Lin 18, Garg-I-Srinivasan 18]

Information-theoretic protocol: via ARE!

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Perfect protocol for addition

For general *f*, computational security via (non-black-box use of) oblivious transfer

[Garg-Srinivasan 18, Benhamouda-Lin 18, Garg-I-Srinivasan 18]

Information-theoretic protocol: via ARE!

Open Questions on Perfect ARE

- OR over any G with perfect security and correctness?
  - Same for XOR over odd $\mathbb{Z}_m$?
- OR/XOR over $\mathbb{N}^{\mathbb{N}}$ with perfect security?
  - One-round shuffle-MPC with perfect security?
- Your favorite f with perfect correctness?

Our Partial Answers (this paper)

- OR over any G with perfect security and correctness? No
  - Same for XOR over odd $\mathbb{Z}_m$? No*

* assuming rational probabilities

- OR/XOR over $\mathbb{N}^{\mathbb{N}}$ with perfect security? No
  - One-round shuffle-MPC with perfect security? No

- Your favorite f with perfect correctness? No******

****** negative results for OR where 0 maps to a low-entropy distribution

OR is complete for perfect correctness [BEGI26]

The Full Answers (epilogue)

- OR over any G with perfect security and correctness? No
  - Same for XOR over odd $\mathbb{Z}_m$? No*
- OR/XOR over $\mathbb{N}^{\mathbb{N}}$ with perfect security? No
  - One-round shuffle-MPC with perfect security? No
- Your favorite f with perfect correctness? No******

The Full Answers (epilogue)

- OR over any G with perfect security and correctness? No
  - Same for XOR over odd $\mathbb{Z}_m$? No

Restriction eliminated by ChatGPT 5.6 Sol

- OR/XOR over $\mathbb{N}^{\mathbb{N}}$ with perfect security? No
  - One-round shuffle-MPC with perfect security? No

- Your favorite f with perfect correctness? No****** Yes!

Positive result by ChatGPT 6 Astra

Ruling Out Perfect ARE for 2-Party OR

Distributions

Fourier transform

$$A_0, A_1, B_0, B_1$$

$$Y_0 \coloneqq A_0 + B_0$$

Perfect security $\quad Y_1 \coloneqq A_1 + B_1 = A_1 + B_0$

Rearranging terms $\quad Y_0 + Y_1 = Y_1 + Y_1$

$$\hat{Y}_0 \odot \hat{Y}_1 = \hat{Y}_1 \odot \hat{Y}_1$$ Convolution Theorem

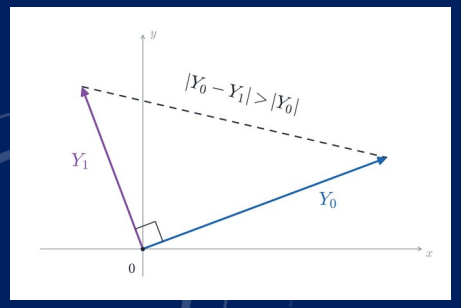
$$\left(\hat{Y}_0 - \hat{Y}_1\right) \odot \hat{Y}_1 = 0$$ Rearranging terms

Impossible for nonzero vectors with disjoint supports (by perfect correctness)

$$\|Y_0 - Y_1\|_2 \leq \|Y_0\|_2$$

$$\left\|\hat{Y}_0 - \hat{Y}_1\right\|_2 \leq \left\|\hat{Y}_0\right\|_2$$

$$\left[\hat{Y}_0 - \hat{Y}_1\right]_i \in \left\{0, \left[\hat{Y}_0\right]_i\right\}$$



Summary

- Main feasibility questions for perfect ARE are settled
- Perfect client-server MPC with honest input-clients still open
  - Perfect ARE is sufficient but may not be necessary
  - At least somewhat Astra-hard (1.5 hours yesterday)

- If we can’t beat AI, let’s join it!